

Deciphering the plain text K1 to K3

This document relates specifically to my current thinking that all the clues for deciphering the fourth cryptographic panel from the [KRYPTOS sculpture](#), installed at the CIA headquarters in Langley, Virginia, come from passages 1 to 3.

Whilst it has been stated many times that Kryptos *can* be broken using only the K4 cipher text, for 33 years, this has been proven to be untrue as the method used to encrypt the panel has never been discovered.

My thinking here is that the method is embedded and explained in panels 1 to 3 and this is my journey of discovery.

Before I move on, for those reading from the Kryptos forum, you may note that my work recently was around the discovery of primes and the speculation of an advanced cipher and key exchange such as Diffie Hellman with Elgamal, before I suddenly switched, almost without warning to examining the plain text in great detail.

In case you're wondering, this was the result of a chain of events that made me question everything I was doing.

1. I was discussing hobbies with a colleague and our conversation turned to adventure game books.
2. Josh LeVeque posted about media quotes related to Kryptos. <https://kryptos.groups.io/g/main/message/23792> This made me realise just how much noise had built up over the years.
3. Rob McCleary posted about the [Greek word ASTATOS](#) which made me realise I'd been totally wrong about LAQUEO, a keyword I'd based much of my previous research around.
4. [Elonka Dunin posted a statement](#) which made me realise just how little we understood about the plain text in K1 to K3.
5. I took a day trip to [Vilanova i la Geltru](#) on the Catalonia coast. On the way back the train broke down and I found myself staring out to sea through a small hole in the cliff face (a tiny breach if you will) whilst thinking about the shape of those 97 characters.

This chain of events then started to come together in my head and brought me directly to this discussion.

Whilst this isn't complete, I'm hoping that this will bring forth a resolution of the cipher text known as K4. It's still 33 years old until November. I'm 44. The

symmetry here would be remarkable. See below.

Traps

For 9 years I have chased fantasies and dreams around K4. I've tried forcing my own will onto the cipher, I've probably made a fool of myself trying to solve it with techniques that do not belong to it and wasted countless hours of my life inventing numbers that do not exist, largely as a result of my own vanity and desire to "Be the one who breaks it using some magical, yet non-existent method of transposing one set of letters into another".

With K4, I think we have all set a trap for ourselves; I certainly fell into a deep one but there are also two very clever ones, one set for us by the sculpture itself, and the second set by our expectations around it.

1. I feel we are geared to over-engineering solutions based on the premise "This must be hard, the CIA, NSA, GCHQ and probably every other intelligence agency in the world can't break it"

Whilst I used to believe this to be true, over the course of a few days in April, my thinking shifted from the need for over-engineering to believing that its complexity comes from its simplicity.

I stated recently in an email to a fellow amateur cryptographer that the bit that makes it hard is that we forget how to think like an artist and infer things that actually are not there.

2. For the second of the two traps; we're all caught up in solving the cipher text yet how many of those who have attempted to decipher KRYPTOS can truly say we've solved the all of the plain text?

My estimation is that most reading this, like myself have decided "K1, Vigenère, Palimpsest, that's cute, move on" and I think this is a real trap that was sprung by the sculpture.

This plain text has a lot more to say.

Avoid K4

Trying to dive straight in to K4 without fully understanding K1 to K3 is probably the biggest trap of all that this sculpture springs on us.

Over the years, many things have been said in retrospect about this cipher and this feeds into the belief that K4 can be solved with some magical combination of other cipher methods.

- Unknown: Can you solve K4 without solving K1 to K3 first?
- JS: Yes but K5 cannot be solved without K4.

People have spent many thousands of combined hours on this cipher. It's one of the greatest cryptographic challenges in the field and whilst it is true that K4 *might* be solvable without K1 to K3, time has proven this to be almost impossible.

Whilst it is hard to completely avoid K4, in theory, this cipher text should be irrelevant to discovering the method used to deciphering it.

Solving K4 should be a matter of putting pieces of the puzzle together. Once the puzzle is assembled, K4 will solve itself.

That said, there will be places throughout this discussion where I reference the K4 cipher text, predominately for correlation between the method and the solution being devised.

Speculation and conjecture

These are yet more traps that are easy to fall into. Whilst both of these are reasonable in context, it is easy to find rabbit holes to go down when trying to look for clues as to how K4 works.

 To serve as an example of this, The numbers in the coordinates given in passage 2 add up to 64. The length of the phrase "SLOWLYDESPARATLYSLOWLY" is 22 characters.

There are 4 passages in the Kryptos sculpture, 97 - 4 is 93. We will come back to these numbers later.

These form the number 93-22-64 which is the bank sort code for the AIB Bank in the town of Althone, which sits on the border of County Roscommon and County Westmeath in the Republic of Ireland.

At the time of writing this, there were 1690 members of the Kryptos forum. In 1691, a dozen men destroyed the bridge over the river Shannon during the second siege of the town; slowing the advance of the Williamite army under General Ginkel. During the siege, 32 heavy cannon and mortar fired one shot every minute at the town, making it the heaviest bombardment of any city in Britain and Ireland up

until that point.

32 is half of 64 and clocks apparently play a large role in this cipher.

The number of Grenadiers Ginkel sent to destroy the Jacobite position was 2000 which is roughly the number of letters carved in the whole sculpture. The letters GG appear twice in the K2 cipher text.

The actions of the 12 men was later captured in the poem A Ballad of Athlone by Aubrey de Vere.

This kind of thinking falls into the realm of confirmation bias and whilst this has absolutely no relevance to Kryptos, it serves to highlight how easy it is to draw completely irrelevant connections between this sculpture and events that have taken place throughout history.

By no means am I saying do not use these techniques. They are useful and inspiration often comes from applying both speculation and conjecture in the right place, at the right time.

During the course of this discussion, I will try to be clear when I believe I am doing either. There is room for both but for every speculative piece of evidence brought to the table, I would expect the legwork to be done to prove it or expect it to be discarded.

Until that point, it should be called out for what it is, and I will try and highlight the ones I believe are relevant.

Keywords

Another trap to fall into is the belief of keywords being discovered inside either the cipher text or the plain text, especially inside K4.

K4 is 97 alphabetic characters with a specific character distribution. Whilst the distribution does not match the standard character distribution, it is only 3 characters less than the number of pieces in a standard English language scrabble game.

The number of possible ways one can rearrange these 97 characters, allowing for substitution where relevant to form new words is approximately $26^{97} \approx 1.31 \times 10^{143}$

This is many orders of magnitude more than the estimated number of atoms in the observable universe $\approx 10^{78} \leq x \leq 10^{82}$

Given that the Oxford English Dictionary contains $\approx 600,000$ words and the Collins English Dictionary contains $\approx 722,000$ words, the chances of finding keywords in this cipher text is extremely high, whilst the chance of finding the exact phrase that forms Kryptos K4 is $1 : 26^{97-1}$

That said, these numbers should not deter you from attempting valid or viable methods of decryption but should instead encourage you to look for the correct method used and not try and brute force the algorithm as so many have tried and failed to achieve over the years.

On PALIMPSEST and ABSCISSA

I would like to talk briefly about these two keywords as they are not only important to solving the related passages 1 and 2 but I believe that they are also important towards passage 4.

The first of these is PALIMPSEST. No, it's not spelt PALIMPCEST, it doesn't have any religious meaning, it's not pointing to a specific scroll; I think we need to accept that the word is literally being given in its natural form and we need a very specific part of the meaning of this word.

Re-Use

The meaning of the word **Palimpsest** is

- A manuscript or piece of writing material on which later writing has been superimposed on effaced earlier writing.
- something **reused or altered** but still bearing visible traces of its earlier form.

The part here that we are looking for is the reused or altered. I think both of these words have meaning here and I will try and demonstrate why I believe this to be the case a little later.

Why this keyword was chosen for passage 1, I believe is because it indicates that passage one will point you towards the section of the sculpture that will be both re-used and altered in order to decipher passage 4.

Coordinates

The word **Abscissa** is chosen for passage 2. I believe that this is because Mr Sandborn is saying "You now have one half of the geometric position". If this is the case, we still have the ordinate to find.

There is another possible meaning to this word as well; in that Mr Sandborn is saying “I’m giving you one half of the cipher”. Whilst this is both a contentious, and speculative reading of the word, it is, I believe still a valid assumption.

Even as I write this, I become aware that there are many more possible meanings for this choice of word and I hope that by the end of this discussion, the true meaning will make sense, if not to myself, at least to you.

On LAQUEO and ASTATOS

Back in 2015, I noted that the misspellings and their counterparts were an anagram of the word LAQUEO. I have spent a considerable amount of time on this word over the years and I fully believe I have fallen into its meaning. In Latin, this word means “A trap” and yes, I think I fell into it. My current belief is that the misspellings of words are a deliberate trap to throw you off the cipher. Think of this in terms of Indiana Jones and a treasure map for the pyramids, littered with traps.

The breakthrough on me recognising this came when a man named Rob McCleary wrote about his discovery of the word Greek word ASTATOS on the Kryptos forum <https://kryptos.groups.io/g/main/message/23883>. This word literally means “Unstable”.

When I read this, I became of the belief that it was telling us that “This half of the cipher is unstable”. To follow through with the treasure map analogy “Here be dragons”.

Just like the word, my thinking has shifted a little in light of more recent discoveries to believe that it also means the cipher is unstable; it shifts and moves; morphs from one thing to another and we should be wary of the form it takes.

That said, just like any good Indiana Jones movie, we need to cross the dangerous part to get to the treasure. For why have a treasure map if it’s supposed to be easy?

Shading

In passage 1, we note the phrase

BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF
IQLUSION

The length of this section is 63 characters.

When I first came to the cipher on 2014, my initial reaction to the word IQLUSION was towards another word. INCLUSION. Although this word is wrong in terms of the cipher text, in an example I give later, it could be construed to carry meaning.

Whilst it is widely agreed upon that the Q in IQLUSION is a deliberate misspelling of ILLUSION, nobody has ever deciphered this phrase; this was also recently pointed out by Elonka Dunin in her reply to the Greek word ASTATOS <https://kryptos.groups.io/g/main/message/23910>.

In terms of another place to look, I am reminded of Sanborn's comment about the K1 plain text. When I asked him where it was from, like a poem or something, he answered that it was an original sentence, written by him, with "carefully chosen wording".

I don't believe we have ever come up with a satisfactory explanation for his comment, could it be related?

It was this exact comment that finally pushed me into what I believe to be the right direction towards finding out exactly what K1 to K3 tell us.

It has to be stated that this caught me off guard by a very wide margin. I always assumed the plain text was well studied and understood. It never occurred to me that nobody had ever really paid it much attention.

Much of it is remarkably obvious once you start looking in detail.

Whilst in the initial stages of the transformation I described in the introduction to this discussion, I still believed this to relate to something hidden in the cipher text, as I broke down more of the plain text, I realised just how wrong we've been all this time.

I am entirely convinced now that Mr Sandborn is pointing us directly to how this cipher works.

Dissection

Consider the phrase, and reduce it down to its relevant components.

BETWEEN something AND something LIES something

Fill in the gaps and you get to

BETWEEN *S – S* AND *A – L* LIES *N – I*

Here, we find 3 character pairs that take you in one direction inside the cipher.

There are two key things to remember about illusions. One, in an illusion anything is POSSIBLE, and the second is that for an illusion to work, it has to be VISIBLE

But:

IT WAS TOT *AL LY IN*VISIBLE HOWS THAT PO *SS* IBLE

Here we find the letters *AL*, *IN* and *SS* but they are in the wrong order.

Place a pivot point between the I and the N and rotate it. When doing so, it is reasonable to consider how when an object orbits another, it often shows the same face.

SS – NI – AL

Axis and Allies

 There is another way to interpret this now. The SS were one of the most brutal forces in the second world war but formed a small part of the AXIS, the collective term for the countries Germany, Italy, and Japan.

The counterpart to that was the ALLIES which was principally made up of Great Britain, The United States of America and Soviet Russia.

In a section of the sculpture where geometry plays a large role, it is hard not to form this connection, especially if you think there can be no good action adventure without WW2 German soldiers running around after you.

Astute observers may note another pair of letters in this text that directly link back to the first.

In passage 1, we're told that BETWEEN *S ... S ...* AND *A ... L ...* **LIES** *N ... I ...*

This is yet another facet of illusions and its existence brings into play another interesting pair. *LY*.

When we then read the first phrase from passage 2, we also note:

IT WAS TOT *ALLY IN*VISIBLE HOWS THAT PO *SS* IBLE

We have an Ally in the SS.

Here we can look to K4 where we find that *SS* appears twice in the cipher separated by the 8 characters *OTWTQSSJQ* which in turn, contains Mr Sanborn's initials *JS* surrounded by the letter *Q*. It may be that this signifies a reversal has taken place and that *JS* has turned and is working for the other side but there are questions about him.

Geo Magnetic Fields

In passage 2, we note the phrase as follows:

THEY USED THE EARTHS MAGNETIC FIELD

In 1906, Bernard Brunhes made the first discovery of Geomagnetic reversal, published in his paper *Les variations de l'aimantation des terres volcaniques de la Chaîne des Puys et de l'Auvergne* [Variations in magnetization of volcanic rocks in the Chaîne des Puys and Auvergne].

This work was later built upon by Motonori Mat(s)uyama in 1929 in the paper "On the magnetization of rocks in Japan.", and again by the English Mathematician and geophysicist Harold Jeffreys in his 1948 paper "The Earth's Magnetic Field and its Secular Variation."

In the early 1950s, a team led by Lloyd V. Berkner discovered that the earths magnetic field was tilted at an angle of approximately 11° from the perpendicular.

⚠ I want to also place a warning to anyone looking to follow this research; stick to the relevant facts. We know Mr Sandborn has an interest in Atomic reaction through his book [Atomic Time](#) and the sculpture [Critical Assembly](#)

Those researching the Geomagnetic field, will undoubtedly come across [Dynamo Theory](#) which will inevitably lead towards [Patrick Blackett](#) who introduced a theory on the Earths magnetic field as a function of rotation, a theory he later contradicted in his paper "The Diurnal Variation of the Earth's Magnetic Field".

My understanding is that Blackett would later go on to be instrumental in aligning Britain with America on the creation of the Atomic bomb.

Whilst this makes for fascinating reading, this is almost certainly irrelevant to the sculpture KRYPTOS.

Don't get too caught up in the speculative, stick to only what you can see.

Key points

1. The magnetic field is tilted 11° off the perpendicular. This **may** be indicative of a Caesar shift somewhere in the cipher. If so, the shift is most likely to be of 11.
2. The magnetic field periodically flips over.

In my understanding, Mr Sandborn has made several references to “Turning” and “Rotating” in relation to K4. Lets keep this in mind.

Intervals

There is another potential element of misdirection at play here. K2 contains:

- $7 \times X$, 5 of which are at the end of statements
- $10 \times Y$
- $2 \times LY$

If we imagine K2 being a conversation between two people, the X symbols indicate a change of speaker when a question is not being asked, or a pause in the conversation which is then followed by a piece of information.

Person 1. IT WAS TOTALLY INVISIBLE HOWS THAT POSSIBLE ?

- this is a lie.

Person 2. THEY USED THE EARTHS MAGNETIC FIELD X

- Quite literally. “THEEARTHSMAGNETICFIELD” is 22 characters long

Person 2. THE INFORMATION WAS GATHERED AND TRANSMITTED UNDERGROUND TO AN UNKNOWN LOCATION X

- Still unsure about this

Person 1. DOES LANGLEY KNOW ABOUT THIS ?

- Still unsure about this

Person 2. THEY SHOULD ITS BURIED OUT THERE SOMEWHERE X

- This is true. It relates to the next passage (K3)

Person 1. WHO KNOWS THE EXACT LOCATION ?

Person 2. ONLY WW THIS WAS HIS LAST MESSAGE X

- This is a lie. Someone else knows

Person 2. THIRTY EIGHT DEGREES FIFTY SEVEN MINUTES SIX POINT FIVE SECONDS NORTH SEVENTY SEVEN DEGREES EIGHT MINUTES FORTY FOUR SECONDS WEST X

- I think this relates to how the string is split up
- This is 110 characters long including non-numeric words
- The sum of the numbers given is 64. This is 33 less than the required 97.
 - if we add to this the 22 from THEEARTHSMAGNETICFIELD, we get 86 leaving 11 to find.

Person 2. LAYER TWO

One thing of interest here is the numbers given as the coordinates.

These numbers add up to 64 $\sum[3, 8, 5, 7, 6, 5, 7, 7, 8, 4, 4] = 64$

The lengths of each word in the phrase “THE EARTHS MAGNETIC FIELD” are $\sum[3, 6, 8, 5] = 22$. This leaves 11 characters to find.

Now, we learn in the next section that 4 characters do not move so only 7 remains.

This gives us a set of numbers $\sum[3, 6, 8, 5, 7] + \sum[3, 8, 5, 7, 6, 5, 7, 7, 8, 4, 4] + 4 = 97$
This is the length of the K4 cipher text.

One thing that is interesting here is the pattern that is forming in the numbers here. I am not entirely sure what to do with this yet but I believe that it relates to the rotation of blocks in the cipher.

Question

What is the purpose of the U to O conversion in this passage?

In Passage 1, the Q indicates that you look between the Qs In passage 3, the A to E conversion is indicative of the missing A in RED, this is backed up by the missing E in DESPARATLY.

I’m missing the meaning of the U in UNDERGRUUND

Thoughts

It is possible, that these indications are telling you when to change direction in the cipher; or a cadence in which to read the letters but both of these options are only conjecture until proven otherwise.

Note the person to whom the Xs belong. Person 1 never contains an X, Person 2 has 5 of them.

One of the reasons I think this, is that in Morse code, an X was occasionally used to indicate a pause or a break between sections.

If that were the case in this message, there should be more Xs (for example between INVISIBLE and HOW, and between WW and THIS); Overall, the beat or cadence of this section feels off.

If this was to be a cadence, then the pattern will almost certainly be 1 – 2 – 2 – 1 – 2 – 1 – 2 – 2 – 2 as this is the order that the speakers appear.

The sum of the numbers in this cadence is 15 = O. Is this sequence incomplete? Is this the meaning of the U-O misspelling?

If this is the case, it could be that the sequence should start with 1 – 2 which would make it a little more symmetrical and might resolve the woodenness of the cadence?

One note that bothers me about the cadence is that it implies a second passage to read from. This might not be a problem as there is another plain text passage of interest that contains all the letters of interest (SS NI AL), in their correct order:

SLOWLY DESPARATLY SLOWLY THE REMAINS OF PA *SS* AGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER AND THEN WIDE *NI* NG THE HOLE *AL* ITTLE I INSERTED THE CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q?

Tutankhamun and the raiders of the missing candle

The whole passage contains

- 1 × X
- 8 × Y
- 4 × LY

Now I come to K3 and the discussion of Howard Carters discovery of the tomb of Tutankhamun. Most of this is conjecture. I think I've not yet fully understood this passage but this tells us a lot.

- SLOWLYDESPARATLYSLOWLY is 22 characters long but it's all LYs so we don't listen to it.
- PRESENTLY is another LY. This is going to take time.

We start with the full passage, but initially set aside the part prior to the *SS* in *PASSAGE*. I am not sure if this is required but this takes time.

Building the machine

In passage 2, we located the boundaries of the room as set out in passage 1. These are: *SS* and *AL*. It is my opinion that within these boundaries, everything we require will fit. Here, I will go through the process of packing the room with all that is required to move forward *possibly into the cipher text known as K4*.

The length of the relevant section is 141 characters. By the time this is finished, you will be left with only 97 characters.

The characters *SS* and *AL* define the boundaries of the room we leave these in place throughout this build.

```
1  SS  AGEDEBRISTHATENCUMBEREDTHELOWERPARTOFTHEDOORW
2      AYWASREMOVEDWITHTREMBLINGHANDSIMADEATINYBREACHI
3      NTHEUPPERLEFTHANDCORNERANDTHENWIDENINGTHEHOLE  AL
```

The first thing we need to do is clear the passage debris, so lets get rid of that.

```
1  SS  THATENCUMBEREDTHELOWERPARTOFTHEDOORWAYWASRE
2      MOVEDWITHTREMBLINGHANDSIMADEATINYBREACHINTH
3      EUPPERLEFTHANDCORNERANDTHENWIDENINGTHEHOLE  AL
```

The next instruction tells us *OF THE DOORWAY WAS REMOVED* so lets get rid of that:

```
1  SS  THATENCUMBEREDTHELOWERPARTWITHT
2      REMBLINGHANDSIMADEATINYBREACHIN
3      THEUPPERLEFTHANDCORNERANDTHENWIDENINGTHEHOLE  AL
```

I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER

OK, lets do that: Take “A TINY BREACH” and move it, overwriting “THATENCUMBE”

```
1  SS  ATINYBREACHREDTHELOWERPARTWITHT
2      REMBLINGHANDSIMADEINTHEUPPERLEF
3      THANDCORNERANDTHENWIDENINGTHEHOLE  AL
```

Next step:

THEN WIDENING THE HOLE A LITTLE I INSERTED THE CANDLE AND PEERED IN

This one confused me until I realised it meant everything after INSERTED so lets do that.

```
1  SS  ATINYBREACHREDTHELOWERPARTWITHT
2      REMBLINGHANDSIMADEINTHEUPPERLEF
3      THANDCORNERTHECANDLEANDPEEREDIN  AL
```

Reading this passage, you will notice that it states “A TINY BREACH RED THE LOWER PART” and THE CANDLE AND PEERED IN should be made in the upper left hand corner.

Note how there is an A missing in the word RE[A]D. If I read this correctly, this passage contains the word DESPARATLY, which uses an A in place of an E, and there is an E missing. I believe the A in READ is the counterpart to the missing E in DESPERATELY.

So, lets do that:

```
1  SS  THECANDLEANDPEEREDINREDTHELOWER
2      PARTWITHTREMBLINGHANDSIMADEINTH
3      EUPPERLEFTHANDCORNERATINYBREACH  AL
```

This, if I’m not mistaken, is a perpetual motion machine. It is constantly reversing itself around TREMBLING HANDS.

THE HOT AIR ESCAPING FROM THE CHAMBER

I believe we do not need this section, it literally escapes.

CAUSED THE FLAME TO FLICKER

I have not yet understood this phrase but it may relate to the change in structure of this phrase as it moves between locations.

BUT PRESENTLY DETAILS OF THE ROOM WITHIN

The word PRESENTLY is a lie but DETAILS OF THE ROOM WITHIN may relate to how K4 should be cut at one end.

EMERGED FROM THE MIST

I believe this is the second cut.

Finally

CAN YOU SEE ANYTHING Q

I'm not the first person to think this is a question directed at the character Q but given that the best place to look for something is from an elevated position, I personally believe that this question is being called up to the Q in IQLUSION who I imagine answering "No, I'm in the dark".

In other words, look between the Qs.

Notes

One thing that is important to note here is that the operation only moves 93 out of the 97 characters. 4 characters (2 at each end), **do not move**

Exchanges

```

1 Rotations:
2 -----
3 THIRTY EIGHT FIFTY SEVEN SIX FIVE SEVENTY SEVEN EIGHT FORTY
  FOUR
4 3, 8, 5, 7, 6, 5, 7, 7, 8, 4, 4,
5 -----
6 Q - L = 17 - 12 = 5
7 U - O = 21 - 15 = 6
8 E - A = 5 - 1 = 4
9 FLAME = 5 TO = 2 FLICKER = 7
10
11 Questions in passage 2: 3
12 Statements in passage 2: 6
13 Questions in passage 3: 1
14 X in passage 2: 7
15 X in passage 3: 1
16 LY in passage 3: 4
17
18
19 passage 2
20 7 X
21 10 Y (one zero = 3, 4)
22 2 LY
23
24 passage 3
25 1 X
26 8 Y
27 4 LY

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```
28 -----  
29 I think these are the exchanges...  
30  
31 DETAILS OF THE ROOM WITHIN = 22  
32 -----  
33 EMERGED FROM THE MIST = 18  
34 -----
```